João Manoel Carvalho

Backend Java Developer

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Backend Java developer with 3 years on a school management platform used by 30+ schools, owning data migrations, query optimization and integrations (facial-recognition turnstiles, Brazil's national school census export, e-signature). Personal projects built with Java 21, Spring Boot and PostgreSQL, backed by integration tests. Open to remote backend Java roles.

SKILLS

Languages: Java (8, 17, 21), Kotlin, SQL, TypeScript, JavaScript.

Backend: Spring Boot, Spring Security, Hibernate/JPA, REST, JWT, Flyway, JSP.

Databases & testing: MariaDB/MySQL, PostgreSQL, JUnit, Testcontainers.

Tools: Git, Docker, GitHub Actions, React.

EXPERIENCE

Junior Java Developer · EDU3 (Brazil) Aug 2023 – Present

Java 8, Hibernate, JSP, Mentawai, MariaDB · school management platform serving 30+ schools

- Migrated 2012–2026 data from a school's legacy system into Edu3: **~1M grades, 9,000 enrollments and 1,500 students**, through a pipeline from CSV backups to SQL staging tables to an import bot, with a rollback routine.
- Took ownership of **search performance across the system**, rewriting Hibernate queries with projections and removing lazy-loading issues; the heaviest searches, including one that used to bring the system down, became much faster.
- Wrote the advanced SQL (MariaDB) the application doesn't cover: **bulk fixes in production** (UPDATE/DELETE with JOINs), management reports, migration source queries and support bug investigations, using subqueries, aggregations, window functions and stored procedures.
- Built a facial-recognition turnstile integration: an on-site Java gateway with a **local queue that keeps recording entries offline**, plus upload of entry photos to Edu3 (in acceptance testing).
- Implemented and validated the 2026 School Census export for INEP (Brazil's education agency) against the official **494-field layout**, checking official domain tables at export time.
- Built per-school billing rules (installments, scholarships, discounts), boleto (Brazilian bank slip) integration and **contract e-signature via Clicksign**.

PROJECTS

fin-mec · github.com/merlinmec/fin-mec · demo: fin-mec-demo.onrender.com/demo

Java 21, Spring Boot, Spring Security, PostgreSQL, Flyway, Testcontainers, React

- Personal finance app built as a modular monolith; balances, credit card statements and forecasts are computed in the backend and **tested against a real PostgreSQL** with Testcontainers.
- **Idempotent** OFX/CSV statement import and recurring transactions: re-running never duplicates data. Includes TOTP 2FA and data deletion under Brazil's privacy law (LGPD).

arvore · github.com/merlinmec/arvore

Java 17, Spring Boot, PostgreSQL, Flyway, JWT, React, GitHub Actions

- Goals and habits app with a JWT-secured API and PostgreSQL integration tests **running in CI on every push**.

EDUCATION & LANGUAGES

B.Sc. in Computer Science · State University of Piauí (UESPI), Brazil 2021 – 2025

Thesis (grade 9.0/10): Transformers vs. classical models for recommender systems (PyTorch, MovieLens).

Languages: Portuguese (native); English C1 Advanced (EF SET 64/100, cert.efset.org/8Zg7F3); Spanish (basic).